

Question ID: 8cbf1415

| Assessment | Test | Domain                            | Skill       | Difficulty                                          |
|------------|------|-----------------------------------|-------------|-----------------------------------------------------|
| SAT        | Math | Problem-Solving and Data Analysis | Percentages | Medium <div><div></div><div></div><div></div></div> |

Question

In a group, **40%** of the items are red. Of all the red items in the group, **30%** also have stripes. What percentage of the items in the group are red with stripes?

Answer

- A. **10%**
- B. **12%**
- C. **70%**
- D. **75%**

Correct Answer: B

Rationale

Choice B is correct. It’s given that in a group, **40%** of the items are red. It follows that the number of red items in the group can be represented by **0.4 $x$** , where  **$x$**  represents the total number of items in the group. It’s also given that of all the red items in the group, **30%** also have stripes. It follows that the number of items in the group that are red and have stripes can be represented by **0.3(0.4 $x$ )**, or **0.12 $x$** . The expression **0.12 $x$**  represents **12%** of  **$x$** . Since  **$x$**  represents the total number of items in the group, it follows that **12%** of the items in the group are red and have stripes.

Choice A is incorrect and may result from subtracting **30%** from **40%** rather than calculating **30%** of **40%**.

Choice C is incorrect and may result from adding **30%** and **40%** rather than calculating **30%** of **40%**.

Choice D is incorrect and may result from calculating the percentage that **30%** is of **40%** rather than calculating **30%** of **40%**.